Graphs, Line Plots, and Shape Sorting
Answer Key

Grade 2–3 Math

Quick Answers

1. 12	4. 12	7. 7	10. 5
2. 31	5. 1, 4, 0, 1	8. —	11. 2
3. 5	6. 4	9. 4	12. 5

Common wrong answers

2. *If they got 26: left one bar out of the sum and added only the three tallest bars*
4. *If they got 10: started counting at the second label and left out the two X marks above 3 inches*
6. *If they got 1: used the length with the tallest stack of X marks as the longest pencil instead of the greatest length on the plot*
7. *If they got 17: added the two bars instead of subtracting to compare them*
11. *If they got 13: found the rectangle's perimeter by adding only one length and one width instead of two of each*

1. Graph A — how many books did Cleo read?

Find Cleo's bar, then slide across to the scale on the left. Cleo's bar stops level with the label 12, and the number printed on top of the bar says 12 too.

$\boxed{12}$

Teaching note: reading a single bar is the one job here — no adding. Students who answer 4 have counted the number of bars instead of the height of one bar.

2. Graph A — how many books altogether?

"Altogether" means add every bar, so all four students must appear in the sum:

$$8 + 5 + 12 + 6 = 31$$

Add in an easy order if it helps: $8 + 12 = 20$, $5 + 6 = 11$, and $20 + 11 = 31$.

$\boxed{31}$

Common wrong answer 26: one bar was skipped (usually Ben's 5, the shortest bar). Ask the student to point at each of the four bars while adding.

3. Line Plot B — how many pencils are 4 inches long?

On a line plot each X stands for one object. Find 4 on the number line and count only the X marks stacked above it: there are 5.

$\boxed{5}$

4. Line Plot B — how many pencils in all?

The total number of pencils is the total number of X marks, so add every stack:

$$2 + 5 + 3 + 1 + 1 = 12$$

$\boxed{12}$

Common wrong answer 10: the two X marks above 3 were left out because counting started at the second label. The first label always has a stack to count too.

5. Shape Set C — sort the shapes by number of sides.

Count the sides of each shape one at a time: P (square) 4, Q (rectangle) 4, R (rhombus) 4, S (trapezoid) 4, T (triangle) 3, U (hexagon) 6.
Now tally them into the chart. Nothing in the set has 5 sides, and a row with nothing in it gets a 0 — an empty row is not the same as a finished row.

$\boxed{\text{3 sides: 1, 4 sides: 4, 5 sides: 0, 6 sides: 1}}$

Check: $1+4+0+1 = 6$, and there are 6 shapes in the set. Every shape must land in exactly one row.

6. Line Plot B — longest pencil minus shortest pencil.

This question is about the *lengths* on the number line, not about how many X marks sit above them. The greatest length that has any X is 7 inches; the least is 3 inches.

$$7 - 3 = 4$$

$\boxed{\text{4 inches}}$

Common wrong answer 1 inch: the tallest stack (at 4 inches) was read as "the longest pencil". Tallest stack = most common length, not greatest length.

7. Graph A — how many more books did Cleo read than Ben?

"How many more" is a comparison, so subtract the shorter bar from the taller one. Cleo's bar is 12 and Ben's is 5.

$$12 - 5 = 7$$

$\boxed{7}$

Common wrong answer 17: the two bars were added. "Altogether" adds; "how many more" subtracts.

8. Shape Set C — why Dev is wrong.

Model answer: Dev is wrong because shapes Q, R and S all have 4 sides and none of them is a square. For example, shape Q is a rectangle: it does have 4 square corners, but its sides are not all the same length — two are long and two are short. A square needs *both* things at once: 4 sides of equal length and 4 square corners. Shape R (the rhombus) shows the other half of the same point: its 4 sides are equal, but its corners are not square.

So "4 sides" only tells you the shape is a quadrilateral. Squares are one special kind of quadrilateral, not the only kind.

Manual review: this is a written explanation, read by a teacher, not machine-checked. Give credit for naming one four-sided non-square from the set AND saying what a square also needs.

9. Line Plot B — the most common pencil length.

The most common value is the one with the tallest stack of X marks. The stacks are 2, 5, 3, 1 and 1 marks tall, and the tallest is the stack of 5, which sits above 4 on the number line. The answer is the *length*, 4 inches — not the height of the stack.

$\boxed{\text{4 inches}}$

10. Line Plot B — how many pencils are longer than 4 inches?

"Longer than 4" means every X mark strictly to the right of 4: the stack above 5, the stack above 6, and the stack above 7. The stack above 4 itself is not included.

$$3 + 1 + 1 = 5$$

$\boxed{5}$

Check: 5 pencils are longer than 4 inches and $5 + 5 + 2 = 12$ pencils in all, which matches problem 4.

11. Square versus rectangle — distance around.

The distance around a shape is its perimeter: add the lengths of all the sides.
Square, 4 equal sides of 6 cm: $4 \times 6 = 24$ cm.
Rectangle, 8 cm long and 3 cm wide — it has *two* lengths and *two* widths: $2 \times (8 + 3) = 2 \times 11 = 22$ cm.
Compare the two: $24 - 22 = 2$.

$\boxed{\text{2 cm}}$

Common wrong answer 13 cm: the rectangle was given only one length and one width ($8 + 3 = 11$), which is half its perimeter. A rectangle has four sides, so both the length and the width are used twice.

12. Synthesis — adding a seventh shape.

(a) The new shape has 4 sides, so it joins the "4 sides" row. That row held 4 shapes (P, Q, R, S), and one more makes

$$4 + 1 = 5$$

$\boxed{5}$

(b) Model answer: Ben is right. There are 7 shapes in all, and half of 7 is $3\frac{1}{2}$. Since 5 is greater than $3\frac{1}{2}$, more than half of the shapes have 4 sides. Another way to say it: 5 shapes have 4 sides and only 2 do not, and 5 is more than 2.

Manual review: part (b) is a written explanation. Part (a) is machine-checked; the reasoning in (b) is read by a teacher. Accept either comparison (5 against $3\frac{1}{2}$, or 5 against 2) as long as the student uses numbers.